1. Momentum

- Linear momentum is a vector quantity, having the same direction as the velocity.
- The units of momentum are Ns

$$Momentum = mass \times velocity \qquad p = mv$$

- **Principle of conservation of linear momentum:** when bodies in a system interact, total momentum remains constant provided no external force acts on the system.
- When both particles A and B move in the same direction

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

- When both particles A and B move towards each other

$$m_A u_A - m_B u_B = m_A v_A + m_B v_B$$

- When both particles stick together, their velocity becomes the same after impact and can be considered as a single particle with a combined mass.

$$m_A u_A + m_B u_B = (m_A + m_B)v$$

{SP20-P04} Question 3:

Three small smooth spheres A, B and C of equal radii and of masses 4 kg, 2 kg and 3 kg respectively, lie in that order in a straight line on a smooth horizontal plane. Initially, B and C are at rest and A is moving towards B with speed $6ms^{-1}$. After the collision with B, sphere A continues to move in the same direction but with speed $2ms^{-1}$

i) Find the speed of B after this collision

ii) Sphere B collides with C. In this collision these two spheres coalesce to form an object D. Find the speed of D after this collision

Solution:

Part (i)

Calculate momentum of system before collision:

$$p = mv$$

$p = (4 \times 6) + 2\,(0) + 3\,(0) = 24\ kgms^{-1}$

Calculate momentum of system after collision:

$$p = (4 \times 2) + 2v$$

Apply conservation of momentum:

$$total\ momentum\ before = \ total\ momentum\ after$$

$24 = 8 + 2v \quad v = \frac{16}{2} = 8.0\ ms^{-1}$

Part (ii)

Calculate momentum of system before collision:

$$p = (8 \times 2) + (3 \times 0) = 16\ kgms^{-1}$$

Calculate momentum of system after collision:

Note: The masses B and C combine to form D

$$p = (2 + 3)\,v$$

$$p = 5v$$

Apply conservation of momentum:

$$16 = 5v$$

$$v = \frac{16}{5} = 3.2\ ms^{-1}$$

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